

EXPERIMENT-13

Implement a Car Price Prediction Model using Python.

Code:

```
import pandas as pd
from google.colab import files
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

print("=" * 70)
print("          CAR PRICE PREDICTION")
print("=" * 70)

# Upload dataset
uploaded = files.upload()

filename = list(uploaded.keys())[0]

# Read dataset
data = pd.read_csv(filename)

print("\nCAR PRICE DATASET")
print("-" * 70)
print(data)

print("\nTotal Number of Cars :", len(data))
```

```
# Convert categorical values into numbers

fuel_encoder = LabelEncoder()
seller_encoder = LabelEncoder()
transmission_encoder = LabelEncoder()

data["Fuel_Type"] = fuel_encoder.fit_transform(data["Fuel_Type"])
data["Seller_Type"] = seller_encoder.fit_transform(data["Seller_Type"])
data["Transmission"] = transmission_encoder.fit_transform(data["Transmission"])

# Input features
X = data[
    [
        "Year",
        "Present_Price",
        "Kms_Driven",
        "Fuel_Type",
        "Seller_Type",
        "Transmission",
        "Owner"
    ]
]

# Target
y = data["Selling_Price"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
```

```

    test_size=0.20,
    random_state=42
)

print("\nTraining Records :", len(X_train))
print("Testing Records :", len(X_test))

# Create Linear Regression model
model = LinearRegression()

print("\nTraining Car Price Prediction Model...")
model.fit(X_train, y_train)

print("Training Completed Successfully.")

# Prediction
prediction = model.predict(X_test)

print("\n" + "=" * 70)
print("          ACTUAL VS PREDICTED PRICES")
print("=" * 70)

for i in range(len(y_test)):
    print(
        "Actual Price =",
        round(y_test.iloc[i], 2),
        " | Predicted Price =",
        round(prediction[i], 2)
    )

```

```
# Evaluation

mse = mean_squared_error(y_test, prediction)
r2 = r2_score(y_test, prediction)

print("\n" + "=" * 70)
print("          MODEL PERFORMANCE")
print("=" * 70)

print("\nMean Squared Error :", round(mse, 3))
print("R2 Score          :", round(r2, 3))
```

```
# New car prediction

print("\n" + "=" * 70)
print("          NEW CAR PREDICTION")
print("=" * 70)
```

```
# Example:

# Year = 2020
# Present Price = 10
# Kms Driven = 15000
# Fuel = Petrol
# Seller = Dealer
# Transmission = Manual
# Owner = 0
```

```
new_car = pd.DataFrame({
    "Year": [2020],
    "Present_Price": [10.0],
    "Kms_Driven": [15000],
    "Fuel_Type": [
```

```

        fuel_encoder.transform(["Petrol"])[0]
    ],
    "Seller_Type": [
        seller_encoder.transform(["Dealer"])[0]
    ],
    "Transmission": [
        transmission_encoder.transform(["Manual"])[0]
    ],
    "Owner": [0]
})

```

```

predicted_price = model.predict(new_car)
print("\nNew Car Details:")
print("Year      : 2020")
print("Present Price : 10.0 Lakhs")
print("Kilometers   : 15000")
print("Fuel Type     : Petrol")
print("Seller Type    : Dealer")
print("Transmission   : Manual")
print("Previous Owners : 0")
print("\nPredicted Selling Price :",
      round(predicted_price[0], 2),
      "Lakhs")
print("\nCar Price Prediction Completed Successfully.")

```

Sample Input:

```

...  =====
      CAR PRICE PREDICTION
      =====
      Choose Files Car_Price_Prediction.csv
      Car_Price_Prediction.csv(text/csv) - 1701 bytes, last modified: 8/10/2026 - 100% done
      Saving Car_Price_Prediction.csv to Car_Price_Prediction.csv

```

Sample Output:

CAR PRICE DATASET							\
...	Car_Name	Year	Present_Price	Selling_Price	Kms_Driven	Fuel_Type	
0	Swift	2018	6.87	5.25	25000	Petrol	
1	City	2017	8.92	6.75	35000	Petrol	
2	i20	2019	7.15	6.10	18000	Petrol	
3	Verna	2018	10.50	7.50	28000	Diesel	
4	Fortuner	2016	32.00	20.50	45000	Diesel	
5	Baleno	2020	8.40	6.80	12000	Petrol	
6	Alto	2017	4.50	2.75	42000	Petrol	
7	WagonR	2018	5.80	3.85	30000	Petrol	
8	Creta	2019	15.00	11.25	22000	Diesel	
9	Nexon	2020	9.80	7.90	15000	Diesel	
10	Ertiga	2017	9.20	6.40	38000	Petrol	
11	Seltos	2020	16.50	13.25	14000	Petrol	
12	Polo	2016	7.10	4.25	50000	Petrol	
13	Rapid	2017	9.50	5.75	42000	Diesel	
14	Amaze	2018	8.75	6.20	32000	Diesel	
15	Ciaz	2019	10.25	7.80	24000	Petrol	
16	Jazz	2017	7.90	5.10	36000	Petrol	
17	Ritz	2015	5.90	3.10	55000	Diesel	
18	Figgo	2016	6.20	3.40	48000	Diesel	
19	Kwid	2020	5.00	3.60	10000	Petrol	
20	XUV500	2017	18.00	11.50	40000	Diesel	
21	Innova	2016	20.00	13.75	52000	Diesel	
22	Thar	2021	15.50	13.80	9000	Diesel	
23	Venue	2020	11.80	9.20	17000	Petrol	
24	Sonet	2021	12.50	10.40	8000	Petrol	
25	Kiger	2021	10.20	8.15	11000	Petrol	
26	Magnite	2020	9.50	7.45	16000	Petrol	
27	Harrier	2019	22.00	17.50	26000	Diesel	
28	Duster	2017	12.00	7.20	39000	Diesel	
29	EcoSport	2018	10.00	6.85	33000	Diesel	

		Seller_Type	Transmission	Owner
...	0	Dealer	Manual	0
	1	Dealer	Manual	0
	2	Dealer	Manual	0
	3	Dealer	Manual	0
	4	Dealer	Automatic	0
	5	Dealer	Manual	0
	6	Individual	Manual	0
	7	Individual	Manual	0
	8	Dealer	Manual	0
	9	Dealer	Manual	0
	10	Dealer	Manual	0
	11	Dealer	Automatic	0
	12	Individual	Manual	1
	13	Dealer	Manual	0
	14	Individual	Manual	0
	15	Dealer	Automatic	0
	16	Individual	Manual	0
	17	Individual	Manual	1
	18	Individual	Manual	1
	19	Individual	Manual	0
	20	Dealer	Manual	0
	21	Dealer	Manual	0
	22	Dealer	Manual	0
	23	Dealer	Manual	0
	24	Dealer	Automatic	0
	25	Individual	Manual	0
	26	Dealer	Manual	0
	27	Dealer	Manual	0
	28	Individual	Manual	0
	29	Dealer	Manual	0

Total Number of Cars : 30

Training Records : 24

Testing Records : 6

*** Training Car Price Prediction Model...
Training Completed Successfully.

=====

ACTUAL VS PREDICTED PRICES

=====

Actual Price = 17.5 | Predicted Price = 16.01
Actual Price = 7.8 | Predicted Price = 7.93
Actual Price = 9.2 | Predicted Price = 9.74
Actual Price = 3.1 | Predicted Price = 2.64
Actual Price = 11.25 | Predicted Price = 11.12
Actual Price = 7.9 | Predicted Price = 8.1

=====

MODEL PERFORMANCE

=====

Mean Squared Error : 0.468
R2 Score : 0.975

=====

NEW CAR PREDICTION

=====

New Car Details:
Year : 2020
Present Price : 10.0 Lakhs
Kilometers : 15000
Fuel Type : Petrol
Seller Type : Dealer
Transmission : Manual
Previous Owners : 0

Predicted Selling Price : 8.46 Lakhs

Car Price Prediction Completed Successfully.